STAFF REVIEW · H446 320 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. MAR holds address; MDR transports data/instruction contents.

2. Invert fixed-width positive representation then add one.

3. Quadratic asymptotic growth.

4. $1+1+1=3=11_2$; sum1 carry1.

I Do — source worked steps

1. Explain the method — Shift mantissa one position left and reduce exponent by one.

We Do — guided CHECK

1. Explain the method
Each midpoint comparison reduces the interval; return index4.

You Do — source mark schemes

1. + PC value 12 copied into MAR=12 [1]; MDR receives ADD 7 from memory location 12 [1]; CIR receives ADD 7 from MDR [1]; PC advances to 13 [1]; +5=00000101 [1], invert11111010 [1], add1 11111011 [2].

2. mid3→7 [1], low4 [1], mid5→11 [1], return5 [1]; base case stops [2], input progresses [1] avoiding infinite stack growth [1].

3. Indicative content: availability across devices, offline access, connectivity, inclusive accessibility, confidentiality, school account permissions, backup and recovery, and cost. Practice marking (authored guidance, not an OCR mark scheme): 7–9 = accurate, sustained evaluation of both options, applied to this scenario with a supported judgement; 4–6 = relevant comparison with some development and a partly justified decision; 1–3 = basic statements, limited context or unsupported decision; 0 = no creditworthy response. Use best-fit and credit equivalent valid arguments.
Model evaluation: Cloud storage gives pupils access to revision materials across school and home devices and helps teachers distribute updates without copying files manually. However, some pupils have unreliable internet or limited access to suitable devices, and any outage can interrupt revision. USB-only storage works without a connection but is easy to lose, can be outdated and may create a safeguarding or data-security risk if sensitive content is included. For an inclusive revision service I would use an access-controlled cloud platform as the main source, while providing approved offline downloads or school-based access where needed. Accessibility settings, restricted permissions, tested backups and a procedure for updating offline copies should be part of the design. The best decision depends on actual pupil connectivity and devices.

Exit Ticket — source answer

1. False. Independent performance and personalised revision evidence matter.